

How We Tested

Test Bed

Processor	Core 2 Duo X6800
Motherboard	Intel D975XB2
Memory	1 GB DDR2 533 MHz
Graphics Card	GeForce 8800GTX
Hard Disk	Seagate SATA 2 400 GB

Test Conditions

1. Zero per cent ambient lighting (completely dark room)
2. No zoom used during testing (except when testing for zoom quality)
3. Projected image size was kept constant throughout the tests
4. Keystone correction was avoided unless absolutely necessary (because it results in loss of quality)
5. DVI connectivity was used. In case of lack of digital connectivity, D-Sub was used
6. Each projector was kept on for 15 minutes more than their rated warm-up time before the pre-test calibration began. This was done to ensure the best possible colour rendition
7. Default settings were not used. We chose to calibrate each projector to get the best visual quality possible

Features We Looked At

1. Common paper specifications like Brightness and Contrast Ratio. While disbelieving paper specs, we went on to check practically in our tests whether each of the test subjects lived up to the claims made
2. Number of preset modes, and keystone correction (and its effectiveness)
3. Connectivity options (such as DVI, D-Sub, S-Video)
4. Quality of optical zoom

5. Weight of the projector (portability is an issue, especially with business-class projectors, while home audiences will look at visual quality rather than size)
6. Control options (inbuilt mouse, remote control with laser pointers, etc.)

The Gamut Of Tests

1. DisplayMate Video Edition: A well-known video evaluation tool. DisplayMate is also used to test monitors of all sorts. We used this software to evaluate visual quality and also to check for any display anomalies. We tested various parameters like contrast, colour accuracy, brightness, moiré, resolution, focusing, streaking, and ghosting.
2. Picture Quality Tests: We used a set of high-resolution bitmap and Photoshop image files. We checked for colour rendition and also gradations in colour. The subject remained constant, but we used varying levels of colour and even greyscale.
3. Presentation Tests: We used a PowerPoint presentation that consisted of text and graphs. We checked primarily for the readability of text, and after that, the reproduction of colours and detail in the graphs.
4. Animation Tests: We used a high-quality DVD copy of *Shrek 2*. This enabled us to verify accurate colour rendition, and the ability to display intricate details. Our non-animated movie was *Alexander*, which contains sufficient matter to accurately judge the projector's ability to provide a satisfying theatre experience.
5. Game Tests: We used *F.E.A.R.* as a benchmark. The game was run at with 4x AF and 4x AA enabled and all visuals set to their maximum values. We used the projectors' native resolutions.

NEC VT590

Jack and master...

The VT590 from NEC is part of their entertainment range. It's an XGA-supporting LCD projector decked in attractive piano white. Build quality is very good, and the body feels sturdy. As opposed to some of the boxy models we received, the VT590 is also curvier, and will suit business and home environments looks-wise.

A fully-functional remote interfaces with the device, and one thing immediately noticeable was the clear demarcations near the connectors at the rear—clearly visible in dim lighting, which is helpful. We've come across a couple of projectors (the BenQ MP611c comes immediately to mind) where we could make out nothing of the legends of each connector unless we turned on our overhead lights. Such minor points add up; manufacturers should think from a consumer's perspective—like the difficulties



in setting up the projector in less-than-ambient lighting.

The VT590 adds a thoughtful on/off switch, something we missed on earlier models. It loses out on Digital Video (DVI) and PC stereo connectivity. Also absent are the USB interface and wireless connectivity.

After calibrating this beauty, we fired up the DisplayMate suite first. We immediately hit resolution problems with the horizontal bar tests. There was a noticeable lack of resolution, and even the solid bar lacked uniformity. The remainder of the test suite passed uneventfully.

Corporate users will be interested to note the lack of detail in text during our presentation tests. This projector is not for you if you plan on using a lot of text applications. These problems persisted in MS Word and even while viewing PDFs. Its not that image quality was bad overall; icons were very decently viewable—it's just the text that appeared somewhat blurry despite repeated attempts at calibration.

The NEC VT590 resurged itself somewhat in the multimedia tests with good performance throughout our test movie as well as in *F.E.A.R.* where the visuals were definitely acceptable.

Priced at Rs 84,000, the NEC VT590 doesn't showcase itself as a strong performer in any one domain. It follows the jack-of-all-trades stereotype and ends up being mediocre at a few tasks and decent at a couple. Not enough for us to recommend it to the serious-minded business or home user.